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On the Physical Properties PEDOT:PSS Thin Films

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ABSTRACT

A complete characterization from optical, morphological, electrical, photo-electrical and thermo-electrical point of view was done for PEDOT:PSS thin films and DMSO and EG sensitized PEDOT:PSS films. The studies on the electrical conductivity and electrical photoconductivity allowed the calculation of different relaxation times. The relaxation time of the electrical conductivity is of order of femtosecond and is multiplied by a factor 10 when PEDOT: PSS thin films were deposited on surfaces sensitized with DMSO and EG. Besides, the photoconduction excitation and relaxation times are of order of seconds. An increase in the relaxation photoconduction time by 2 was observed for films deposited on surfaces sensitized with DMSO and by 1.2 for films deposited on surfaces sensitized with EG. The time response of electrical conductivity after exposure to light adds supplementary knowledge for the understanding of the inertial processes, or hysteresis behavior of organic or perovskite solar cells involving PEDOT:PSS films. The electrical conductivity increases with the temperature and the use of DMSO and EG as surfactants lead to higher values of electrical conductivity and Seebeck coefficient. A better stability of the electrical conductivity with the temperature increase, was noticed for films deposited on surfaces sensitized with DMSO.

1. Introduction

PEDOT:PSS (3,4-ethylenedioxythiophene): poly(styrenesulfonate) thin films have attracted a high interest because of their applicability in the field of organic electronics and new generation solar cells thanks to their low sheet resistance, optical transparency and easy processability. Present, PEDOT-PSS is used as buffer layer or as hole transporter in organic light–emitting diodes (OLED), organic (OPV) or perovskite (PVK) cells solar cells [1–11], [12–17]. The classical architecture of the most common organic and perovskite solar cells involving PEDOT:PSS layers are: ITO/PEDOT:PSS/Polymer:Fullerene/LiF/Al and ITO/PEDOT:PSS/$CH_3NH_3PbI_3$/PCBM/BCP/Au respectively. The technological development is however slowed down by the difficulty to obtain reproducible and long term stable results. A lot of factors influence the devices performances such as: diversity of characteristics and quality of the precursor's products, the large number of parameters involved in thin films depositions and manufacturing processes [18–28]. Moreover, the complexity of the characterisation methods [29], [30] is enhanced by the fact that most of these devices are not stable in very short scale time (seconds, minutes or hours), or are highly influenced by the ambient conditions (light, temperature, humidity etc.) [31–33].

Even if much progress has been made in the understanding of the processes inside these materials for optoelectronic devices, some characteristics are still not enough known or insufficiently taken into account in the explanation of the behaviour of photovoltaic devices. Commonly, the explanation of the characteristics of the new generation solar cells based on organic or perovskite materials is mainly based on the properties of the active materials composing these devices. Many effects as: the lateral (or parallel to surface) conduction of PEDOT:PSS, the photoconduction, the variation of the electrical conductivity with temperature etc. are neglected in the solar cells evaluation. However, all changes in the electrical resistivity of the PEDOT:PSS layers during the evaluation of the efficiencies of solar cells will influence the final results. The time elapsed between the moment of exposure to light and the moment at which the current characteristic is traced can also influence the estimated efficiencies and can lead to very different results. These observations need to be related to the time of response of the constitutive materials to different excitation factors. In our previous studies on the electrical properties stability of individual conductive polymers (PEDOT:PSS, P3HT and P3HT:PCBM) we noticed that the electrical properties during many cycles of heating and cooling, in dark and after exposure to light are remarkably stable and reversible [34], [35].

The lateral conduction and the photoconduction can contribute

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positively on the solar cells efficiency by increasing of the number of photo-generated carriers, which can then contribute to the increase of the energy conversion efficiency. Many of these parameters are combined and these combined properties are extremely interesting because they can be exploited simultaneously to produce electricity, both, by photo- and thermo-electric effects.

For more than one decade, the interest of the use of PEDOT:PSS layers in electronic and optoelectronic organic devices was pointed on, due to the fact that it conducts to an increase of the open circuit voltage of the organic solar cells, reduces the short circuits risks or it increases the life time and external quantum efficiencies of OLED. More recently the thermo-electrical properties of PEDOT-PSS were pointed out too [36–40]. A high electrical conductivity of PEDOT:PSS thin films is a key point. Its value can vary in function of the producer or formulations with many orders of magnitude. It was also shown that, by introducing ethylene glycol (EG) or organic polar solvents such as dimethylsulfoxide (DMSO) or dimethylformamide (DMF) in the aqueous solutions of PEDOT-PSS, the electrical conductivity of spin coated thin films can be significantly increased [41–43]. It was shown that by introducing DMSO in the PEDOT:PSS solutions, DMSO interacts with the sulfonic groups of PSS and this causes the PEDOT polymer chain to rearrange . After an optimal doping with DMSO, the conductivity of PEDOT:PSS decreases because of the excess removal of PSS resulting in longer average distance between the conductive PEDOT grains. The morphological models and the influence of co-solvents DMSO and/or EG are given in [44], [45]. From these models, one can observed that the presence of these solvents in the deposition solution lead to the formation of islands domains, in which the polymer chains are well arranged. At the interfaces between these domains the order is disrupted. In order to keep the benefit of the DMSO effect on the arrangement of polymer chains and at the same time avoid the effect of the formation of islands, in this paper EG and DMSO were first deposited on the whole substrate surface, as thin layers, prior to the deposition of the PEDOT:PSS thin films. Hence, the influence of surfactant acts on the whole surface of the PEDOT:PSS films in the same way.

In this study we report on a complete characterization on optical, electrical, opto-electrical and thermo-electrical properties of spin-coated PEDOT-PSS thin films and DMSO and EG sensitized PEDOT:PSS thin films, in order to better understand the charge transport and the time response to different excitations factors such as temperature or light irradiation. The relaxation time of the electrical conductivity, and the excitation and relaxation times of photoconductivity, were determined by ellipsometry and by direct measurements, respectively.

These time response analyses could also later lead to a better understanding of the OPV and PVK solar cells photo-response.

2. Experimental section

Highly conductive PEDOT with negatively charged poly(styrenesulfonate), PSS- as counter ion was purchased from Ossila (M122 – PH1000) in the form of an aqueous dispersion. Dimethylsulfoxyde and ethylene glycol were purchased from VWR and used as surfactant agents for PEDOT: PSS thin films preparation. The molecular structures of these compounds are depicted in Fig. 1.

A DMSO or an EG thin film was deposited prior to the PEDOT: PSS deposition layer. Surface sensitization with surfactants and PEDOT-PSS thin films deposition were done by spin-coating on glass substrates and on ITO patterned thin films, using the same spinner speeds program as following: step 1 - 1000RPM (10 s), step 2 - 1500 RPM (20 s), step 3 - 1000 RPM (10 s), step 4 – 0 RPM (5 s). After spin-coating, the samples were dried in on a hot plate at 110 °C during 30 min.

Films thickness measurements were done by profilometry using a Dektak profilometer and by ellipsometry using a Horiba Jobin Yvon ellipsometer. The samples deposition parameters and thin films thickness are given in Table 1.

Samples morphology was visualised by contact mode Atomic Force Microscopy (AFM) using a Thermomicroscope Autoprobe LP Research.

The investigations of the optical properties were made by spectroscopic ellipsometry (SE) technique using a UVVISELTM ellipsometer from Horiba Jobin Yvon, with a 75 W high discharge Xe lamp in the spectral range from 260 nm to 2100 nm. All the spectra were recorded at room temperature, at an incident angle of 70°. The configuration chosen for the modulator (M), analyzer (A) and polarizer (P) positions were: M = 0°, A = 45° respectively. After the relatively fast recording of Ψ and Δ spectra, the next step was the construction of an appropriate optical model for the samples in order to fit the experimental values. These investigations were joined with the recorded absorption/transmission spectra of a double beam spectrophotometer LAMBDA 950 UV/Vis/NIR spectrophotometer. The transmittance spectra were recorded with unpolarized light, at room temperature, in the wavelength range 260-1100 nm.

The electrical conductivity measurements were using gold and /or ITO electrodes during one cycle of heating and cooling, between 25-120 °C. Here, the studies were made under ambient light (5 W/m^2) and under a white light source irradiation (250 W/m^2).

The thermo-electrical properties of these films were determined using the system described in Fig. 2. The open-circuit thermo-voltage (ΔV) was measured between two gold probes placed on the sample at a distance of 7 mm, using a Keithley 2182A nanovoltmeter. The temperature gradient (ΔT) was realized using two Peltier elements, controlled by two programmable power supplies and measured by K-type digital thermocouples.

3. Results and discussion

3.1. Surface topography

DMSO and EG interacts with the sulfonic groups of PSS and separates it by weakening the attraction between PEDOT and PSS chains [44]. This can cause the PEDOT polymer chains to rearrange themselves and these modifications can be observed in the surface topography by a reduction of the surface roughness. The morphological and structural properties of PEDOT:PSS thin films and consequently their electrical properties, can strongly influence the solar cells efficiencies [46] because the properties of the conducting polymer, are determinant for the electrical conduction mechanism through the whole solar cell structure. The RA and RMS averages roughness values are given in Table 2. As one can see the use of surfactants on the substrates, prior to PEDOT: PSS deposition, lead systematically to a decrease of the films roughness. The lowest value of the roughness is obtained when EG is used as surfactant. We also remark a noticeable difference between the values of roughness for films deposited on glass compared to films deposited on ITO. From an electrical point of view, as will be seen later, the electrical mobility and the electrical conductivity of DMSO and EG sensitized PEDOT:PSS thin films are higher than those of PEDOT:PSS films.

3.2. Optical properties

The transmission spectra are given in Fig. 3. The use of surfactants (without dry between the two deposition steps of PEDOT:PSS films) leads to an increase of the transmission of PEDOT:PSS films with about 5% both for DMSO and EG.

The measured ellipsometric spectra for films deposited on glass were fitted by minimizing the squared difference χ2 between the measured and calculated values of the ellipsometric parameters I_s and I_c using the DeltaPsi2 software of Horiba Jobin Yvon. I_s and I_c are given by:

$$I_s = \sin 2\psi \sin \Delta \quad (1)$$

$$I_c = \sin 2\psi \cos \Delta \quad (2)$$

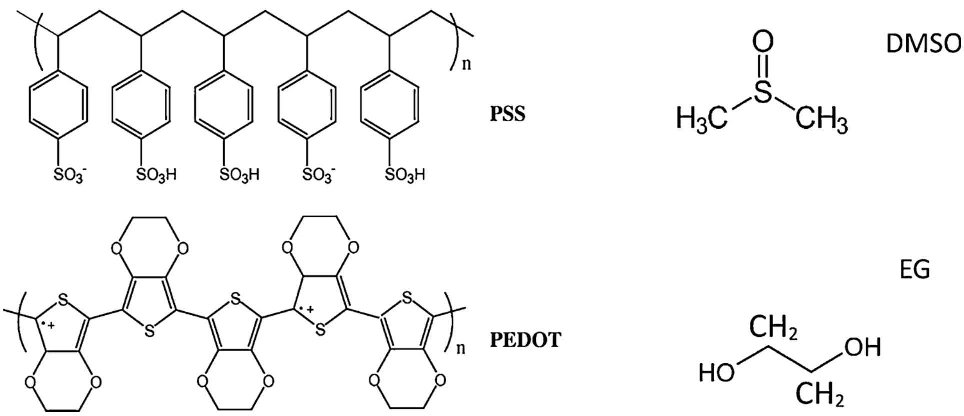

Fig. 1. Molecular structures for PEDOT:PSS, DMSO and EG.

and

$$\chi^2 = \left[\frac{1}{(2N-P)}\right]\sum_i^N \left[(I_{si}^{exp} - I_{si}^{cal})^2 + (I_{ci}^{exp} - I_{ci}^{cal})^2\right] \tag{3}$$

where N is the number of data points and P the number of model parameters [47]. Generally fits are considered enough good if $\chi^2 < 10$.

The DeltaPsi2 software allows to calculate, the "pseudo" dielectric functions, real and imaginary part, $< \varepsilon 1 >$ and $< \varepsilon 2 >$ respectively, as well as the "global" refractive index and the "global" extinction coefficient of the samples including the substrate and the layer(s).

$$\tilde{\varepsilon} = <\varepsilon_1> + i <\varepsilon_2> = \tilde{n}^2 = (n+ik)^2 = \sin(\theta)^2\left[1 + tan(\theta)^2\left(\frac{1-\tan(\psi)e^{i\Delta}}{1+\tan(\psi)e^{i\Delta}}\right)^2\right] \tag{4}$$

The direct measurement of $< \varepsilon 1 >$, and $< \varepsilon 2 >$, or n-exp (global) and k-exp (global) on the basis of Eq. (4) can be compared with the theoretical values obtained by introducing for the individual layers the known dispersions models of different materials from the data basis of the DeltaPsi2 software, and good fits between the measured n-exp and k-exp and calculated n (n-fit) and k (k-fit) can be obtained, for more than one dispersion model. The individual refractive index, n, individual extinction coefficient, k, or individual dielectric functions (real and imaginary part) for each individual layers should then be extracted from the model on the basis of the dispersion formula which gives the best fit. Our models were optimised according to the procedure described in our previous paper and the best fits were obtained using, for the PEDOT:PSS layer, an anisotropic model based on the sum of the single and double Lorentz and Drude oscillators (classical) dispersion formula [48]

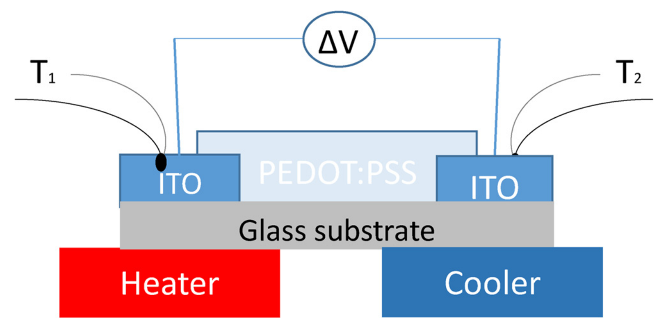

Fig. 2. Thermoelectric measurement setup.

Table 1
Sample deposition parameters.

No	Sample	Preparation spin coating steps	Sample thickness (nm) from profilometry measurements
P1	PEDOT: PSS (undoped)	1000/1500/1000/0 (RPM) 10/20/10/5 (s) Heat treatment 100 °C – 30 min	60-80
P2	DMSO/PEDOT:PSS	First layer DMSO 1000/1500/1000/0 (RPM) 10/20/10/5 (s) Second layer PEDOT:PSS 1000/1500/1000/0 (RPM) 10/20/10/5 (s) Heat treatment 100 °C – 30 min	60-80
P3	DMSO*/PEDOT:PSS	First layer DMSO 1000/1500/1000/0 (RPM) 10/20/10/5 ***Dry – 10 min*** Second layer PEDOT:PSS 1000/1500/1000/0 (RPM) 10/20/10/5 (s) Heat treatment 100 °C – 30 min	100-120
P4	EG/PEDOT:PSS	First layer EG 1000/1500/1000/0 (RPM) 10/20/10/5 (s) Second layer PEDOT:PSS 1000/1500/1000/0 (RPM) 10/20/10/5 (s) Heat treatment 100 °C – 30 min	100-120

Table 2
RA and RMS roughness values from AFM micrographs.

Sample	RA (nm)		RMS (nm)	
	on glass	on ITO	on glass	on ITO
PEDOT: PSS (undoped)	10.3	1.9	13.9	2.9
DMSO/PEDOT:PSS	7.4	1.8	9.6	2.8
DMSO*/PEDOT:PSS	9	2	12.5	2.8
EG/PEDOT:PSS	5.3	1.6	7	2.4

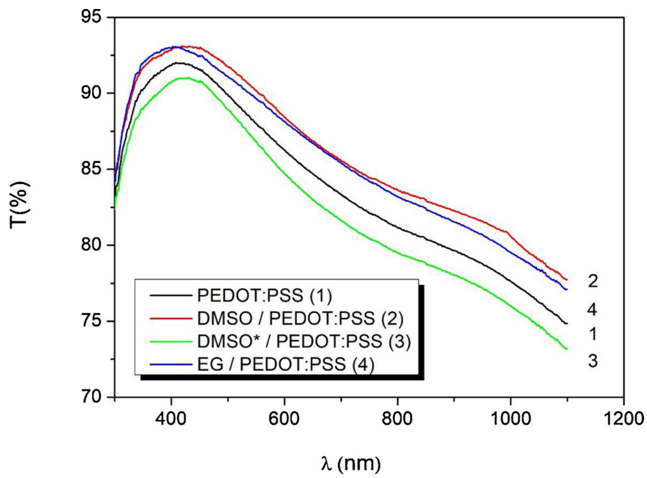

Fig. 3. Transmission spectra for PEDOT: PSS, DMSO and EG sensitized PEDOT:PSS thin films on glass.

$$\varepsilon = \varepsilon_\infty + \frac{(\varepsilon_s - \varepsilon_\infty)\omega_t^2}{\omega_t^2 - \omega^2 + i\Gamma_0\omega} + \frac{\omega_p^2}{-\omega^2 + i\Gamma_D\omega} + \sum_{j=1}^{2}\left(\frac{f_j\omega_0^2 j}{\omega_0^2 j - \omega^2 + i\gamma_j\omega}\right) \tag{5}$$

In this equation, ε, ε_∞, f_1 and f_2 are parameters without dimension and ω, ω_p, Γ_D, Γ_0, γ_1 and, γ_2 are expressed in eV.

Fig. 4a and Fig. 4b depict the experimental and best fitted spectra of pseudo-dielectric functions < ε1 > , and < ε2 > respectively, and Fig. 5 and Fig. 6 give the calculated values for the individual PEDOT: PSS layers for the ordinary and extraordinary optical axis respectively. The insets of Fig. 4a and Fig. 4b depict the ellipsometric models used for calculations. The thickness of the films calculated from the ellipsometric models are given in Table 3. The calculated values are in agreement with the direct measured values obtained by profilometry.

According to Drude's theory in metals, the plasma frequency is given by:

$$\omega_p^2 = \frac{4\pi ne^2}{\varepsilon_0\varepsilon_\infty m_e^*} \tag{6}$$

where **n** is the carriers concentration, $\boldsymbol{m_e^*}$ is the effective mass of charge carriers, $\boldsymbol{\varepsilon_\infty}$ and $\boldsymbol{\varepsilon_0}$ represents the dielectric constants of medium and free space [49]. On the other hand, the electrical conductivity in this model is given by:

$$\sigma = \frac{ne^2\langle\tau\rangle}{m_e^*} \tag{7}$$

where $\langle\tau\rangle$ is the mean value of the electrical conduction relaxation time, which means the mean values of time between two successive collisions. Hence from the Eqs. (6) and (7) it results that:

$$\langle\tau\rangle = \frac{4\pi\sigma}{\varepsilon_0\varepsilon_\infty\omega_p^2} \tag{8}$$

The values of the electrical conductivity relaxation time on the basis of the electrical measurements and ellipsometric models are given in

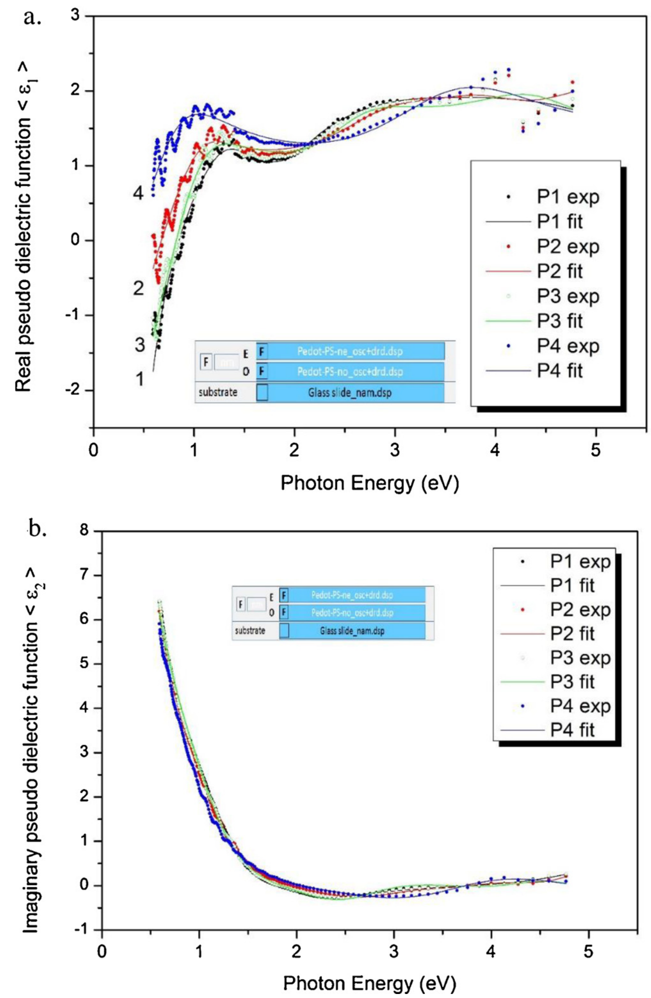

Fig. 4. Real (Fig. 4a) and imaginary (Fig. 4b) pseudo dielectric function for the studied samples on glass substrates: P1: PEDOT:PSS thin films on glass, P2: PEDOT:PSS thin film sensitized with DMSO on glass substrate, P3: PEDOT:PSS sensitized with DMSO, then dry, P4: PEDOT:PSS thin films sensitized with EG. Dots represent the ellipsometry experimental data points and the continuous line to the best fitting ellipsometry model.

Table 4. We notice that the use of wet surfactants (samples noted by DMSO/PEDOT: PSS and EG/PEDOT:PSS), conducts to an increase by ten of the electrical conductivity relaxation time (70 fs, compared to about 7 fs without surfactant). From semiconductors physics theory the relaxation time is directly related to the charges carriers mobility [50], [51], hence the increase of the relaxation time is equivalent to an increase of charges carriers mobility and this observation is coherent with a re-arrangement of polymer chains. As an example, the relaxation time for Copper is of 27 fs [52].

The experimental values obtained for the pseudo dielectric functions (Fig. 5) are similar to the ones obtained by Laskarakis et al. [53]. However, different from [53] in which the best modelling was obtaining with an isotropic model, our best fits were obtained by considering an anisotropic PEDOT:PSS layer, in accordance with different experimental observations in which is mentioned that PEDOT:PSS present an anisotropy [53].

From the calculations using this ellipsometric model, we extracted the real and imaginary dielectric functions for the individual PEDOT: PSS layers, for the two optical axis, ordinary Fig. 5 and extraordinary Fig. 6, respectively. The graphs obtained for the ordinary dielectric functions are in agreement with the results obtained by Gasirovski et al. which consider an isotropic model too. We observe the same behaviour with the use of surfactants, even if the sample preparation procedure is

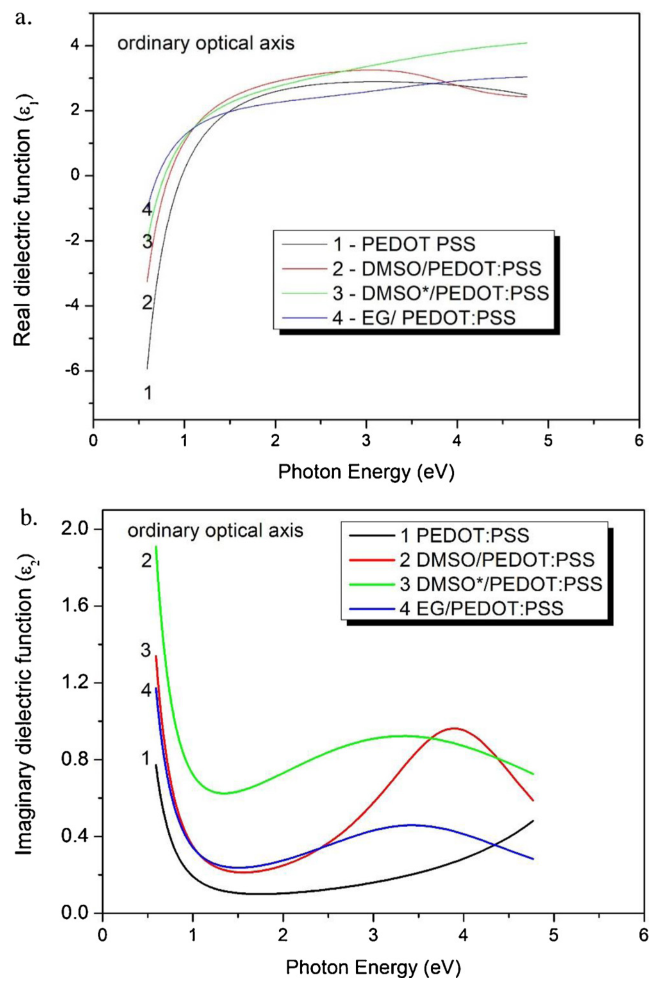

Fig. 5. Individual real (Fig. (5a) and imaginary (Fig. 5b) dielectric function for the ordinary optical axis for the different PEDOT:PSS films extracted from the best fitting models.

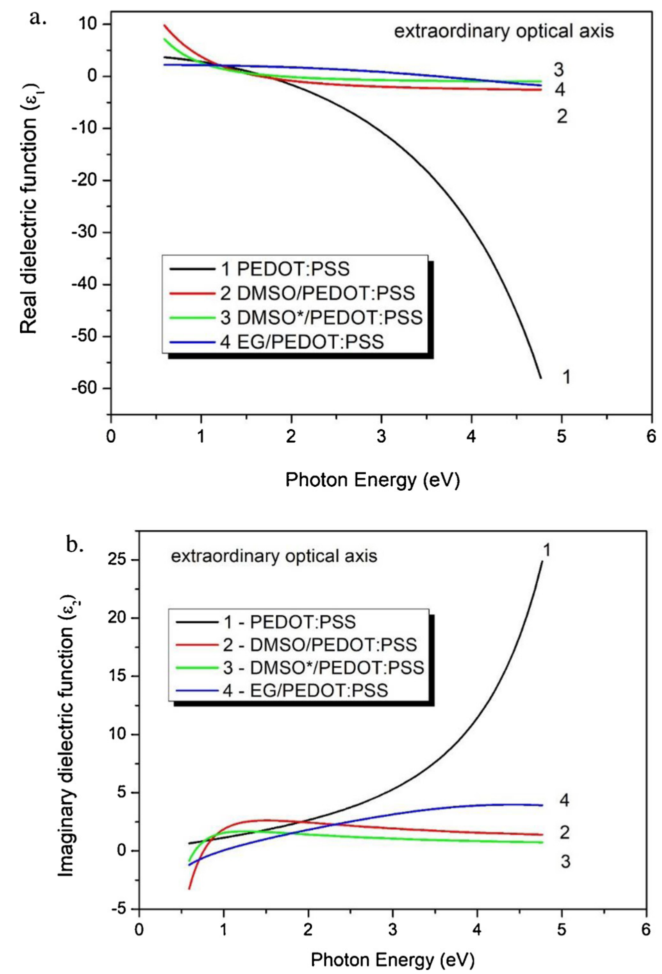

Fig. 6. Individual real (Fig. 6a) and imaginary (Fig. 6b) dielectric function for the extraordinary optical axis for the different PEDOT:PSS films extracted from the best fitting models.

different in our case, namely, a shift of the $< \varepsilon 1 >$ pseudo dielectric function values towards smaller photon energy values and an increase of $< \varepsilon 2 >$ values when surfactants are used [42].

From our anisotropic model, a supplementary information and an evident behaviour modification in the presence of surfactants, is remarked from the calculation of dielectric functions in the direction of the extraordinary optical axis (Fig. 6). Indeed the presence of surfactants conduct to a completely different response of the samples in the domain of photons with high energies (2-5 eV, 240-600 nm), proving once again, by this indirect optical technique, a morphological modification of the PEDOT: PSS polymers films in the presence of surfactants.

3.3. Electrical Conductivity

When masks are used for the measurement and calculation of solar cells efficiency, one can indeed consider that the parallel electrical conductivity of the PEDOT: PSS is not important. But, in reality, (such is the case of modules), the whole surface is irradiated, and not only the area above the back electrode so the lateral electrical conduction and photoconduction contribute positively to the device efficiency and need to be taken into account (See Fig. 7). The measurements of the electrical conductivity for PEDOT: PSS thin films and DMSO and EG sensitized PEDOT-PSS films at room temperature are in agreement with the observation of other authors, for samples obtained by mixing the surfactants with the PEDOT: PSS aqueous solution [54]. We remark an increase of the electrical conductivity when surfactants are used. For the sample DMSO*/PEDOT: PSS, the intermediary step (dry) prior to the PEDOT-PSS film is not recommended because it conducts to a decrease of the electrical conductivity.

The studies of the variation of the electrical conductivity with the

Table 3
Samples thickness calculations from ellipsometry.

Sample	Thickness from ellipsometry measurements (nm)	Error (nm)	χ^2
PEDOT: PSS (undoped)	75	± 5	1.51
DMSO/PEDOT:PSS	76	± 2	2.23
DMSO*/PEDOT:PSS	110	± 5	2.66
EG / PEDOT:PSS	100	± 5	2.54

Table 4
Conductivity relaxation time values resulted from calculations on the basis of Drude Model Eqs. (6–8). $\sigma_{(el)}$ – electrical conductivity values from direct measurements, ω_p - plasma wavelengths, ε_∞ - dielectric constant calculated theoretical values from ellipsometric models (Eq. (8)).

Sample	$\sigma_{(el)}$ (Ω^{-1} cm^{-1})	$\omega_{p(o)}$ (eV)	$\varepsilon_{\infty(o)}$	$< \tau >$ (fs)
PEDOT:PSS	54	1.821	4.942	7.972
DMSO/PEDOT:PSS	208	1.553	2.887	72.282
DMSO*/PEDOT:PSS	13	1.398	4.067	3.957
EG/PEDOT:PSS	116	1.172	2.963	68.972

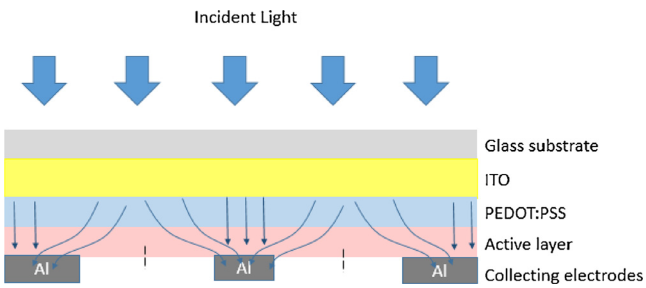

Fig. 7. Collecting charge paths to electrodes in the architecture of solar cells using PEDOT:PSS layers.

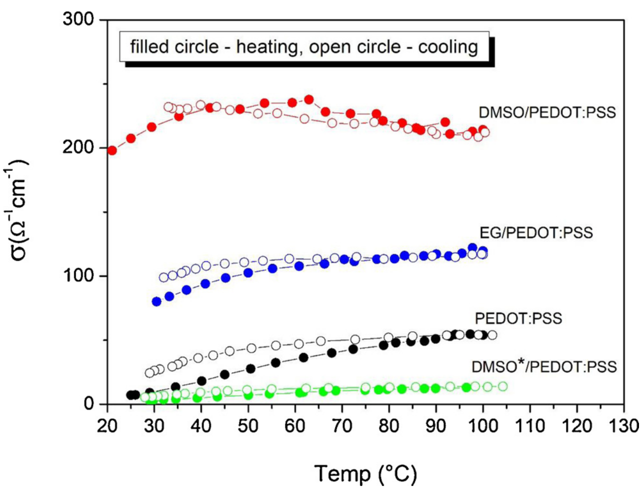

Fig. 8. The variation of the electrical conductivity of different PEDOT:PSS films in function of the temperature and under ambient illumination ($\Phi = 0.5\ \mathrm{mW/cm^2}$) and in open atmosphere during one cycle of heating and cooling between 25 and 105 °C.

temperature for films with or without surfactants (Fig. 8) show a behavior similar to the semiconductors, namely an increase of the conductivity with the increase of the temperature. After the first heating, the polymers suffers modification since during the first cooling the values of the electrical conductivity are systematically higher than those during heating at the same temperature. The use of DMSO as surfactant conduct to PEDOT:PSS films with higher conductivity and less sensitivity (higher stability) of the electrical conductivity with the temperature in the range 25 to 110 °C.

3.4. Photoconductivity

To study the photoconductivity response, the samples were subjected to intermittent irradiation with a white light source ($250\ \mathrm{W/m^2}$), light "on" during 15 minutes followed by measurement of 15 minutes after the light was switch off. Assuming the theoretical modeling equations of the photoconductivity in semiconducting materials, at the exposure to light, the difference ($\Delta\sigma$) in electrical conductivity due to the photo-generated charges follow the Eq. (9), and after the interruption of the light irradiation the difference of ($\Delta\sigma$) electrical conductivity follow the Eq. (10).

$$\Delta\sigma(t) = \Delta\sigma_{st}[1 - \exp(-t/\tau)] \tag{9}$$

$$\Delta\sigma(t) = \Delta\sigma_{st}\exp(-t/\tau) \tag{10}$$

$\Delta\sigma_{st}$ – represents the difference between the value of the conductivity in saturated regime and the electrical conductivity in dark. In equation (9), τ - represents the excitation time and in equation (10), τ-represents the relaxation time. As it results from the photoconductivity curves (Fig. 9 and Fig. 10) analysis the use of surfactants definitively modifies

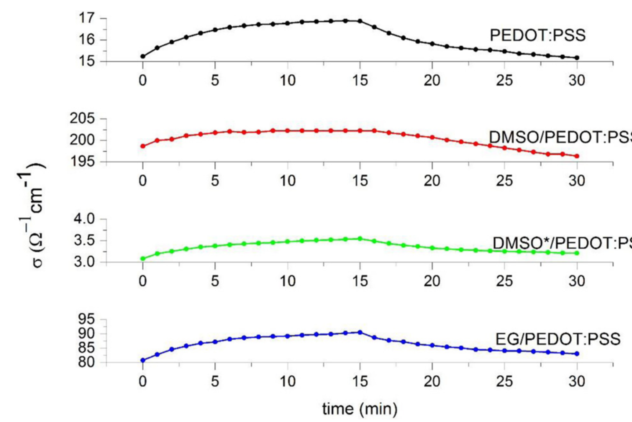

Fig. 9. The variation of the electrical conductivity after the exposure to a white light source ($\Phi = 25\ \mathrm{mW/cm^2}$) during 15 minutes and after switching off the light during 15 minutes (from minute 15 to minute 30).

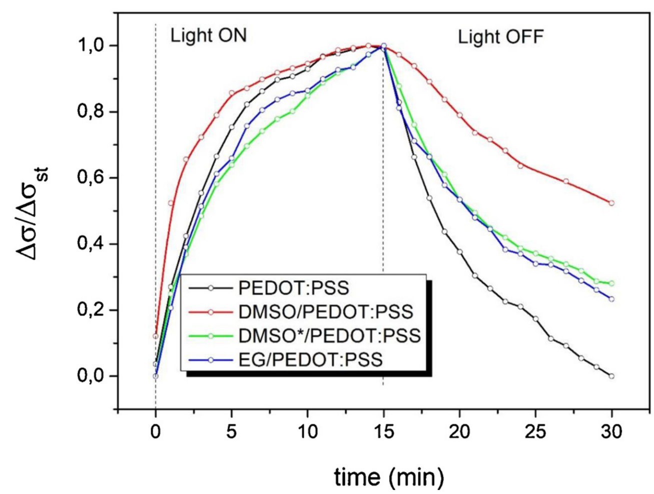

Fig. 10. The variation of the photoconductivity after the exposure to a white light source ($\Phi = 25\ \mathrm{mW/cm^2}$) during 15 minutes and after switching off the light during 15 minutes.

Table 5
Photoconductivity relaxation time calculated from the fitting curves of the experimental data on the basis of Eqs. (9 and 10).

Sample	Excitation time (light on) τ_{on} (s)	Relaxation time (light off) τ_{off} (s)
PEDOT:PSS	234	361
DMSO/PEDOT:PSS	143	635
DMSO*/PEDOT:PSS	322	324
EG/PEDOT:PSS	261	430

the kinetics of photo-generated charge carriers (see Table 5). The photoconductivity excitation time decreases when DMSO is used as surfactant and the relaxation time increases, which means that the life time of photo-generated carriers after irradiation is much longer in the case of DMSO. The same tendency is observed when EG is used as surfactant. The modification of the electrical conductivity of PEDOT:PSS with a saturation after about 15 minutes, can partly explain the instabilities observed in the I-V curves, which describe the photo-response of organic and hybrid or perovskite solar cells using PEDOT:PSS in the first minutes of characterization.

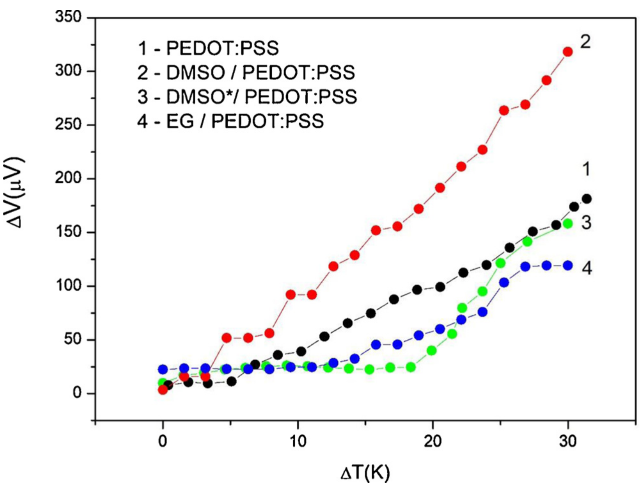

Fig. 11. The thermoelectric voltage in function of the difference of the temperature measured in a planar configuration.

3.5. Thermoelectric properties

In order to complete the physical properties characterization of these films, the thermo-electrical properties were also studied. The maximum thermo-electrical voltage obtained was about 0.3 mV for a difference of temperature of 30 degrees, corresponding to a thermoelectric power of about 30μV/K, (Fig. 11). The highest thermo-electrical voltage was obtained for DMSO sensitized PEDOT:PSS films and these values correspond to the usual values obtained for these materials [55], [56].

4. Conclusions

A complete characterization from optical, morphological and electrical point of view was done for PEDOT:PSS thin films and DMSO and EG sensitized PEDOT:PSS films. Direct measured properties of the material and those calculated from ellipsometric modelling were compared. The ellipsometry models allowed to have a more deeply information concerning the influence of surfactants and the modification of the optical properties in the direction of the extraordinary optical axis. The studies on the electrical conductivity and electrical phtotoconductivity allowed the calculation of different relaxation time. These time-depending experiments allow to better explain the behavior of these films used in OPV and PKV solar cells and can complete the understanding of data and experimental observations concerning the solar cells behavior. The relaxation time of the electrical conduction is of order of femtoseconds and is multiplied by 10, when the surfaces were sensitized with DMSO and EG, prior to PEDOT:PSS thin films deposition. The photoconduction excitation times are of order of seconds. The relaxation photoconduction time also increases by using surfactants, the increase factors being of about 2 for DMSO and about 1.2 for EG. The increase of relaxation times are equivalent to an increase of charge carriers mobility. The use of surfactants clearly conducts to an improvement of the electrical and optical properties of PEDOT:PSS by a rearrangement of the polymer chain in some preferential directions, as it results from the significant difference remarked in the dielectrical functions for films without and with surfactants. The use of DMSO as surfactant leads to higher values of electrical conductivity and Seebeck coefficient, compared to EG.

Declaration of Competing Interests

None.

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